OM JEE

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PROJECTS

Telehealth & Real-Time Consultation Engine

Node.js, Express.js, React.js, MongoDB, Socket.io, JWT, Redis, bcryptjs

- **Real-Time Communication:** Built a bi-directional messaging engine with Socket.io, cutting latency for live doctor-patient consultations compared to HTTP polling.
- **Guest Access Pipeline:** Designed a short-lived, stateful JWT guest-login flow enabling emergency consultation access without full registration.
- **Automated Data Lifecycle:** Built background worker scripts to periodically purge inactive guest profiles and stale chat history, improving storage efficiency.
- **Presence Tracking:** Implemented real-time online/offline status and typing indicators by managing active socket connection state in-memory.
- **Secure Auth & RBAC:** Used bcryptjs for password hashing and JWT for stateless role-based access control across Admin, Doctor, and Patient roles.

Distributed URL Shortener & Analytics Engine

Node.js, Express.js, MySQL, Redis, Docker, Base62 Encoding

- **Stateless Backend:** Built a horizontally scalable Node.js/Express.js backend supporting high-throughput concurrent traffic with no local session dependency.
- **Distributed ID Generation:** Implemented collision-free short URLs using Redis atomic INCR for global IDs, encoded with a custom Base62 algorithm.
- **High-Availability Caching:** Designed a cache-aside pattern with Redis for sub-millisecond redirects, with fail-open fallback to MySQL on cache outage.
- **Horizontal Sharding:** Implemented hash-based sharding across MySQL instances, routing by short-code hash to avoid cross-shard joins.
- **Rate Limiting:** Deployed an IP-based sliding-window rate limiter backed by Redis to secure the write path against abuse.

TECHNICAL SKILLS

- **Languages:** C++, C, Python, JavaScript
- **Web & Frameworks:** Node.js, Express.js, React.js, REST APIs, HTML5, CSS3
- **Databases & Caching:** MySQL, MongoDB, Redis
- **Real-Time & Auth:** Socket.io (WebSockets), JWT, bcryptjs, Role-Based Access Control
- **Tools & DevOps:** Docker, Git, GitHub
- **Core CS:** Data Structures & Algorithms, OOP, DBMS, Computer Networks, Operating Systems

EDUCATION

The National Institute of Engineering (NIE), Mysuru *Sept 2023 - Jul 2027*

B.E. in Information Science & Engineering | CGPA: 8.2

ACHIEVEMENTS & CERTIFICATIONS

- **Data Structures & Algorithms:** Solved 150+ problems on LeetCode and GeeksforGeeks, building consistent problem-solving proficiency.